Served stats

Throughput, latency and quality of each served recipe, by model and hardware

2026-10-20 18:44 UTC

Rendered 2026-10-20 18:44 UTC from stats/*.json, the ledger every harness run and harness export adds to (python3 -m harness stats renders it again). Aggregate tok/s, greedy, 256-token replies, the recipe's own ruler; engine rounds from the server's own counters.

Overview: by model and hardware

Aggregate tok/s as prose/code/structured/list; first token in seconds (one stream / the most streams).

model	hardware	one stream	most streams: total	first token (s)	suite	measured
DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	55/100/62/71	4: 106/162/112/179	0.19 / 0.73-0.99		2026-10-18
DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	55/104/63/71	8: 135/203/144/203	0.19 / 1.35-1.84		2026-10-18
DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	53/98/61/69	4: 106/158/111/176	0.20 / 0.75-1.00		2026-10-18
DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	55/101/63/71	4: 104/156/109/173	0.19 / 0.41-0.64		2026-10-17
DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw abliterated	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	65/95/87/95	8: 226/319/250/359	0.10 / 1.76-3.96		2026-10-20
DeepSeek-V4.1-Flash-1017, 2/3-bit experts	1x Apple M5 Ultra (256 GB unified), macOS	71/105/88/104	4: 127/138/132/146	0.10 / 0.43-0.58	tools 58/60, json 30/30, gsm8k 245/250, ifeval 132/150	2026-10-20
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	42/60/57/58		0.17	json 30/30, ifeval 103/150	2026-10-17
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	27/48/39/33	8: 78/115/98/110	0.18 / 1.49-1.83		2026-10-17
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	-/-/-/-	16: 107/120/119/130	/ 2.69-3.38		2026-10-18
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	-/-/-/-	16: 106/120/131/138	/ 1.32-1.62		2026-10-18
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	-/-/-/-	16: 117/134/148/159	/ 1.34-1.66		2026-10-18

model	hardware	one stream	most streams: total	first token (s)	suite	measured
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE				json 30/30, ifeval 103/150	2026-10-17
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	28/38/33/31	16: 124/156/150/145	0.18 / 8.50-17.20		2026-10-16
GLM-5.3-Flash EXL3 4bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	28/39/34/31	16: 86/101/100/102	0.18 / 3.23-3.72		2026-10-17
GLM-5.3-Flash, EXL3 4 bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	46/69/57/59		0.18	tools 59/60, json 30/30, gsm8k 246/250, humaneval 159/164, ifeval 118/150	2026-10-14
GLM-5.3-Flash, MLX 4-bit	1x Apple M5 Ultra (256 GB unified), macOS	74/78/78/86		0.11	tools 57/60, json 30/30, gsm8k 246/250	2026-10-16
Gemma 4 12B, MLX 4-bit + MTP assistant	1x Apple M5 Ultra (256 GB unified), macOS	140/223/205/248		0.04	tools 58/60, json 30/30, gsm8k 239/250	2026-10-16
Gemma 4 12B, oQ8e (8-bit) + MTP assistant	1x Apple M5 Ultra (256 GB unified), macOS	93/142/146/158	4: 262/324/328/340	0.05 / 0.12-0.12		2026-10-16
Gemma 4 31B, MLX 4-bit + DFlash	1x Apple M5 Ultra (256 GB unified), macOS	87/158/157/130		0.08		2026-10-16
Gemma 4 31B, oQ8e (8-bit) + DFlash	1x Apple M5 Ultra (256 GB unified), macOS	40/63/52/55	4: 110/173/142/180	0.10 / 0.24-0.27		2026-10-16
NVIDIA Nemotron 3.5 Bolt 74B-A9B (24 GB: IQ2_XXS experts, Q8_0, NVFP4 Mamba) + vision	1x NVIDIA RTX 5090 (32 GB)	150/150/150/150		0.03	tools 58/60, json 30/30, gsm8k 241/250, humaneval 153/164, ifeval 120/150	2026-10-15
Ornith-1.5-122B, 4-bit MLX affine (g64)	1x Apple M5 Ultra (256 GB unified), macOS	66/67/66/67	4: 176/178/178/180	0.08 / 0.25-0.35		2026-10-20
Ornith-1.5-122B, 4-bit MLX affine (g64)	1x Apple M5 Ultra (256 GB unified), macOS	85/85/85/85	4: 219/224/238/227	0.05 / 0.12-0.16		2026-10-20
Qwen3.8-27B, MLX 4-bit + DFlash2	1x NVIDIA RTX 5090 (32 GB)	241/459/310/314		0.10		2026-10-16
Qwen3.8-Flash-Next INT4-Mixed (AutoRound)	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	78/127/102/130	16: 304/516/392/525	0.03 / 0.45-0.54		2026-10-17
Qwen3.8-Flash-Next, 6-bit with vision	1x Apple M5 Ultra (256 GB unified), macOS	142/210/173/206	2: 198/262/222/272	0.11 / 0.17-0.19		2026-10-17
Qwen3.8-Flash-Next, MLX 6-bit	1x Apple M5 Ultra (256 GB unified), macOS	145/206/172/194	4: 566/816/673/854	0.09 / 1.63-3.07	gsm8k 242/250, ifeval 134/150	2026-10-11
Step-5-Preview, MLX 2-bit experts (8-bit rest) + MTP	1x Apple M5 Ultra (256 GB unified), macOS	38/50/40/48		0.24		2026-10-16

DeepSeek-V4.1-Flash-1017, 2/3-bit experts on 1x Apple M5 Ultra (256 GB unified), macOS

Target deepseek-v41-m5: DeepSeek-V4.1-Flash-1017, 2/3-bit experts with images and DSpark drafts, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as DeepSeek-V4.1-Flash. Latest: run 20261020-092851-dsv41-quality-off (2026-10-20 09:28, plan dsv41-quality-off, done).

Serving: engine_commit 8557236a; binary_sha256 bc0e5d67944f9a1d9f043d1489af4be990646e3ae512e277d0f4772d67cdd572; served_model {"object": "list", "data": [{"id": "DeepSeek-V4.1-Flash-Abliterated", "object": "model", ''; checkpoint {'repo': 'deepseek-ai/DeepSeek-V4.1-Flash-1017', 'expert_bits': 2, 'layer_bits': '10-29:3'; chip Apple M5 Ultra; 274877906944; suite_commit 277c842a1481

Suite

check	passed
drafts	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2
max-tokens	3/3
tools	58/60
json	30/30
ifeval	132/150
gsm8k	245/250
mgsm	219/240
mmlu	303/342
math500	459/500

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 93%; total = 3072.00M us

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-20 09:28	dsv41-quality-off	done				1488/1614

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-20 09:19	dsv41-speed	done	104.9			
2026-10-20 08:25	dsv41-speed	done	87.7			
2026-10-20 06:19	dsv41-speed	done	86.4			
2026-10-20 04:35	dsv41-quality-off	done				1490/1614
2026-10-20 04:30	dsv41-speed	done	88.5			
2026-10-19 09:08	dsv41-speed	done	90.5			
2026-10-18 20:28	dsv41-quality	done				1487/1614
2026-10-18 18:44	speed	done	87.5			
2026-10-18 17:24	speed	done	92.5			
2026-10-17 19:33	speed	done	96.3			
2026-10-17 17:54	speed	done	90.5			
2026-10-17 14:57	speed	done	56.7			

DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw abliterated on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP

Target dsv41-1017-ablit: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw abliterated, TensorFold native (Zig) TP=2 on two DGX Sparks, 4 slots, 1M window. Engine: TensorFold native (Zig, CUDA) + EXL3 TP=2 kit, recipe defaults (replay prefill, DSpark drafts, thinking on); served as DeepSeek-V4.1-Flash-1017. Latest: run 20261020-131445-dsv41-speed8 (2026-10-20 13:14, plan dsv41-speed8, done).

Serving: image d9acaf504779; patches 8eb5919ff0a4; ablit 0bdc2036; served {"object": "list", "data": [{"id": "DeepSeek-V4.1-Flash-1017", "object": "model", "created; pack 6c4679e6407806ab; suite_commit 35b24fedcc37

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	65.3 (65.3)	94.6 (94.6)	87.3 (87.3)	94.9 (94.9)	0.10-0.16 s
2	91.9 (45.9)	132.3 (66.2)	109.1 (54.6)	163.3 (81.6)	0.15-0.24 s
4	119.5 (29.9)	164.3 (40.0)	136.1 (34.3)	211.2 (51.4)	0.25-0.41 s
8	226.2 (27.9)	318.9 (41.1)	249.9 (30.6)	358.8 (43.0)	1.76-3.96 s

Checks

check	result
memory-idle	measured: head 6.5 GiB available; worker 8.4 GiB available

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-20 13:14	dsv41-speed8	done	94.6			
2026-10-20 13:05	quick	done	94.3			114/114
2026-10-20 12:55	quick	done	94.1			114/114

DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10-native-vision: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw on native TensorFold (main + jayleaton's kit engine 9131e10a ported onto it), native images and video, TP=2 on two DGX Sparks, 4 slots, pool 200K. Engine: TensorFold native (Zig, CUDA): main 9ccf4727 + the DeepSeek-V4.1 kit engine, vision tower (TF_DSV41_IMAGES=native), video as sampled frames, DSpark drafts; served as DeepSeek-V4.1-Flash-TF. Latest: run 20261018-032401-dsv41-speed (2026-10-18 03:24, plan dsv41-speed, done).

Serving: image 7b3d42e0e5865d34; pack 778baeaf9fbcd351

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	55.2 (55.2)	100.4 (100.4)	62.5 (62.5)	71.4 (71.4)	0.19-0.26 s
2	81.0 (40.5)	124.9 (62.5)	84.2 (42.1)	143.6 (71.8)	0.31-0.40 s
4	106.5 (26.4)	161.6 (40.0)	111.9 (26.9)	178.7 (43.1)	0.73-0.99 s

Checks

check	result
memory-idle	measured: head 6.6 GiB available

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-18 03:24	dsv41-speed	done	100.4			

DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10-native-vision8: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw on native TensorFold (main + the kit engine), native images and video, TP=2 on two DGX Sparks, 8 slots, pool 400K, prefill-dip reserve 2 GiB. Engine: TensorFold native (Zig, CUDA): main 9ccf4727 + the DeepSeek-V4.1 kit engine, vision tower, video as frames, DSpark drafts, 8 slots; served as DeepSeek-V4.1-Flash-TF. Latest: run 20261018-092218-dsv41-speed8 (2026-10-18 09:22, plan dsv41-speed8, done).

Serving: image 7b3d42e0e5865d34; pack 778baeaf9fbcd351

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	55.4 (55.4)	103.5 (103.5)	62.7 (62.7)	70.9 (70.9)	0.19-0.26 s
2	81.5 (40.7)	126.2 (63.1)	89.4 (44.7)	142.7 (71.4)	0.30-0.48 s
4	108.8 (26.8)	163.8 (40.3)	113.7 (28.4)	180.4 (44.4)	0.71-0.85 s
8	134.8 (16.7)	202.7 (25.4)	143.6 (17.4)	203.3 (24.8)	1.35-1.84 s

Checks

check	result
memory-idle	measured: head 6.3 GiB available

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-18 09:22	dsv41-speed8	done	103.5			

DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10-native: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw on native TensorFold (main + the deepseek_v41 CUDA family, PRs 667-727), TP=2 on two DGX Sparks, 4 slots, pool 900K. Engine: TensorFold native (Zig, CUDA): main 9ccf4727 + deepseek_v41 series, kit serving profile, DSpark drafts, images as placeholder; served as DeepSeek-V4.1-Flash-TF. Latest: run 20261018-020350-dsv41-speed (2026-10-18 02:03, plan dsv41-speed, done).

Serving: image 00b298dda4527ad6; pack 778baeaf9fbcd351

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	53.3 (53.3)	98.1 (98.1)	61.0 (61.0)	69.3 (69.3)	0.20-0.28 s
2	78.4 (39.2)	121.8 (60.9)	82.5 (41.3)	139.0 (69.5)	0.30-0.42 s
4	105.7 (26.2)	158.5 (39.4)	110.6 (26.5)	176.4 (41.4)	0.75-1.00 s

Checks

check	result
memory-idle	measured: head 6.3 GiB available

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-18 02:03	dsv41-speed	done	98.1			

DeepSeek-V4.1-Flash-1017 EXL3 2.9bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over Ro

Target dsv41-1017-gb10: DeepSeek-V4.1-Flash-1017 EXL3 2.9 bpw, jayleaton's DeepSeek-V4.1 recipe (Python, TensorFold 0.6.0 + patches), TP=2 on two DGX Sparks, 4 slots. Engine: TensorFold 0.6.0 (Python) + jayleaton's DeepSeek-V4.1 patches, recipe defaults (replay prefill, DSpark drafts); served as DeepSeek-V4.1-Flash-TF. Latest: run 20261017-231853-dsv41-speed (2026-10-17 23:18, plan dsv41-speed, done).

Serving: image eb2f599de2fb618a; pack 778baeaf9fbcd351

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	54.7 (54.7)	100.6 (100.6)	62.6 (62.6)	71.3 (71.3)	0.19-0.26 s
2	77.4 (38.7)	126.5 (63.3)	84.0 (42.0)	142.6 (71.3)	0.30-0.39 s
4	103.5 (25.4)	155.8 (38.4)	109.0 (27.7)	173.2 (41.0)	0.41-0.64 s

Checks

check	result
memory-idle	measured: head 8.1 GiB available

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 23:18	dsv41-speed	done	100.6			

Gemma 4 12B, MLX 4-bit + MTP assistant on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-12b-m5: Gemma 4 12B 4-bit (groups of 64), MTP assistant drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-12b. Latest: run 20261016-113616-g12-quality (2026-10-16 11:36, plan g12-quality, done).

Serving: engine_commit 74005ebe; binary_sha256 215b4b7e3cd14a9d8178b4dc885d9f1dffbebd0f4ec2e9b274653820b71da8d2; served_model {"object": "list", "data": [{"id": "gemma4-12b", "object": "model", "created": 1792168585,; chip Apple M5 Ultra; 274877906944

Suite

check	passed
tools	58/60
json	30/30
gsm8k	239/250
mmlu	264/342

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 11:36	g12-quality	done				591/682
2026-10-16 11:31	quick	done	223.3			112/114

Gemma 4 12B, oQ8e (8-bit) + MTP assistant on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-12b-oq8e-m5: Gemma 4 12B oQ8e (8-bit, groups of 64), MTP assistant drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-12b-oq8e. Latest: run 20261016-175459-speed (2026-10-16 17:54, plan speed, done).

Serving: engine_commit d9e36b7f; binary_sha256 2e29fee5da96ec1ffb1ba41302ac949dc12205572154f21b800823450eb4b2b0; served_model {"object": "list", "data": [{"id": "gemma4-12b-oq8e", "object": "model", "created": 179219; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	92.8 (92.8)	142.3 (142.3)	145.7 (145.7)	158.1 (158.1)	0.05-0.05 s
4	261.5 (64.8)	324.2 (81.3)	328.5 (80.7)	340.0 (83.6)	0.12-0.12 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.55	5.525	47%	26.35	12
4	2.944	3.177	62%	37.94	49

Suite

check	passed
drafts	12/12
concurrent	12/12

Checks

check	result
structured	pass: 18/18 obey; a bad regex -> 400

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 17:54	speed	done	142.3			24/24
2026-10-16 17:37	speed	done	141.2			24/24

Gemma 4 31B, MLX 4-bit + DFlash on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-31b-m5: Gemma 4 31B 4-bit (groups of 64), DFlash drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-31b. Latest: run 20261016-085420-quick (2026-10-16 08:54, plan quick, done).

Serving: engine_commit e8b60549; binary_sha256 d90e44c9818187ae4bc61d06312989cd796033b5dea6e0fc02164fe2730660fe; served_model {"object": "list", "data": [{"id": "gemma4-31b", "object": "model", "created": 1792158874,; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	87.1 (87.1)	157.9 (157.9)	156.7 (156.7)	130.4 (130.4)	0.08-0.10 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.288	13.003	18%	24.71	12

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	92.1 (92.1)	161.4 (161.4)	146.6 (146.6)	126.1 (126.1)	0.09-0.10 s

Engine rounds (ruler-after):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.333	13.215	18%	25.34	17

Suite

check	passed
soak	60/60
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3

check	passed
thinking	10/10
copy	2/2
max-tokens	3/3

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 75%; total = 3072.00M us
structured	pass: 18/18 obey; a bad regex -> 400
sampling	pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors
turns	pass: resumed == again == cold: True; cached 5494 of 5564; ttft resumed 0.137s, cold 0.137s

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 08:54	quick	done	157.9			114/114
2026-10-16 06:28	quick	done	156.5			114/114
2026-10-15 20:09	quick	done	153.8			114/114

Gemma 4 31B, oQ8e (8-bit) + DFlash on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-31b-oq8e-m5: Gemma 4 31B oQ8e (8-bit, groups of 64), DFlash drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-31b-oq8e. Latest: run 20261016-180201-speed (2026-10-16 18:02, plan speed, done).

Serving: engine_commit d9e36b7f; binary_sha256 2e29fee5da96ec1ffb1ba41302ac949dc12205572154f21b800823450eb4b2b0; served_model {"object": "list", "data": [{"id": "gemma4-31b-oq8e", "object": "model", "created": 179219; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	40.3 (40.3)	63.1 (63.1)	52.4 (52.4)	54.8 (54.8)	0.10-0.11 s
4	109.7 (27.7)	172.9 (44.8)	141.5 (34.2)	180.3 (44.3)	0.24-0.27 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	2.759	10.847	16%	52.41	12
4	2.384	3.645	38%	63.19	49

Suite

check	passed
drafts	12/12
concurrent	12/12

Checks

check	result
structured	pass: 18/18 obey; a bad regex -> 400

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 18:02	speed	done	63.1			24/24

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-best: GLM-5.3-Flash EXL3 4bpw, families/glm with 4-bit dense / 8-bit head, DFlash2 and 16 streams (PRs 855+979+984 and packing), TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm, packed linears (4/8, bf16 freed), DFlash2, 16 streams, prompts together (prefill_many); served as GLM-5.3-Flash-EXL3. Latest: run 20261017-195233-glm-ab-ifeval (2026-10-17 19:52, plan glm-ab-ifeval, done).

Serving: binary cf50867102a60c38

Suite

check	passed
ifeval	103/150
json	30/30

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 19:52	glm-ab-ifeval	done				133/180
2026-10-17 16:52	glm-quality	done	59.6			348/400
2026-10-17 16:26	glm-streams	done	59.5	131.3	1.27-1.66 s	24/24
2026-10-17 15:56	glm-streams	done	67.7	132.8	1.23-1.59 s	15/24
2026-10-17 15:28	glm-streams	done	68.3	127.5	2.63-3.30 s	24/24
2026-10-17 14:38	glm-streams	done	68.7	123.7	1.77-2.24 s	24/24
2026-10-17 12:40	glm-streams	done	69.4	125.0	2.55-3.17 s	24/24
2026-10-17 11:31	glm-streams	done	69.8	136.4	2.55-3.14 s	24/24

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-dflash2: GLM-5.3-Flash EXL3 4bpw, families/glm with DFlash2 drafts (--drafter, 4-bit), TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm + DFlash2 (incoai/GLM-5.3-Flash-DFlash2, opt-in); served as GLM-5.3-Flash-EXL3. Latest: run 20261017-003221-glm-streams8 (2026-10-17 00:32, plan glm-streams8, done).

Serving: binary 511e7e0521628504

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	27.3 (27.3)	48.4 (48.4)	39.2 (39.2)	33.1 (33.1)	0.18-0.24 s
4	60.0 (14.7)	84.4 (20.8)	75.6 (19.2)	88.1 (21.6)	0.71-0.88 s
8	78.1 (9.7)	115.4 (14.3)	98.4 (12.4)	109.8 (12.9)	1.49-1.83 s

Suite-together

check	passed
concurrent	12/12
drafts	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 00:32	glm-streams8	done	48.4			24/24

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-main: TensorFold main 76aafcf2 (with #855, #979, #984, #995 merged): GLM-5.3-Flash EXL3 4bpw, DFlash2, 16 streams, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm (main); served as GLM-5.3-Flash-EXL3. Latest: run 20261018-131205-glm-qual1014 (2026-10-18 13:12, plan glm-qual1014, done).

Serving: binary e2169dae62202aec

Ruler-c1-16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	42.8 (42.3)	67.2 (67.4)	55.3 (54.8)	53.1 (52.9)	0.16-0.22 s

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	90.0 (11.2)	120.8 (15.1)	114.7 (14.1)	115.5 (14.1)	1.23-1.57 s
16	107.2 (6.6)	119.6 (7.3)	119.4 (7.3)	129.6 (8.1)	2.69-3.38 s

Suite

check	passed
concurrent	12/12
drafts	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-18 13:12	glm-qual1014	done		119.6	2.69-3.38 s	24/24

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-many-mtp: #1014 merged onto TensorFold main 76aafcf2, MTP head only (no --drafter), GLM-5.3-Flash EXL3 4bpw, TP=2. Engine: TensorFold native, CUDA: families/glm (many-mtp); served as GLM-5.3-Flash-EXL3. Latest: run 20261018-205115-glm-qual1014 (2026-10-18 20:51, plan glm-qual1014, done).

Serving: binary 4f1a478269c0e4fa

Ruler-c1-16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	42.1 (41.9)	52.5 (52.2)	50.2 (49.9)	49.0 (49.2)	0.17-0.22 s

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	74.8 (9.3)	95.5 (11.9)	100.9 (12.6)	106.0 (13.2)	0.84-1.05 s
16	105.9 (6.6)	119.7 (7.3)	131.4 (8.3)	138.3 (8.5)	1.32-1.62 s

Suite

check	passed
concurrent	12/12
drafts	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-18 20:51	glm-qual1014	done		119.7	1.32-1.62 s	24/24
2026-10-18 14:04	glm-qual1014	done		118.6	1.28-1.58 s	24/24

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-many: #1014 merged onto TensorFold main 76aafcf2: batched prompts (prefill_many) with DFlash2's feed and the hold fix, GLM-5.3-Flash EXL3 4bpw, TP=2. Engine: TensorFold native, CUDA: families/glm (many); served as GLM-5.3-Flash-EXL3. Latest: run 20261018-202804-glm-qual1014 (2026-10-18 20:28, plan glm-qual1014, done).

Serving: binary 4f1a478269c0e4fa

Ruler-c1-16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	42.5 (42.6)	67.6 (68.2)	56.1 (55.9)	55.8 (53.2)	0.17-0.23 s

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	92.4 (11.6)	131.0 (16.2)	125.4 (15.6)	147.8 (17.1)	0.82-1.03 s
16	117.4 (7.3)	134.1 (8.1)	147.9 (9.3)	158.6 (9.7)	1.34-1.66 s

Suite

check	passed
concurrent	12/12
drafts	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-18 20:28	glm-qual1014	done		134.1	1.34-1.66 s	24/24
2026-10-18 14:29	glm-quality	done	60.1			1947/2418
2026-10-18 13:40	glm-qual1014	done		134.1	1.35-1.67 s	24/24

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-packed: GLM-5.3-Flash EXL3 4bpw, families/glm with 4-bit dense / 8-bit head, DFlash2 and 16 streams (PRs 855+979+984 and packing), TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm, packed linears (4/8, bf16 freed), DFlash2, 16 streams, prompts one at a time; served as GLM-5.3-Flash-EXL3. Latest: run 20261017-195233-glm-ab-ifeval (2026-10-17 19:52, plan glm-ab-ifeval, done).

Serving: binary cf50867102a60c38

Suite

check	passed
ifeval	103/150
json	30/30

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 19:52	glm-ab-ifeval	done				133/180

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-port: GLM-5.3-Flash EXL3 4bpw, main's families/glm with PR 855's port, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm on 1.0.12 (PR 855's port); served as GLM-5.3-Flash-EXL3. Latest: run 20261016-213802-glm-streams (2026-10-16 21:38, plan glm-streams, done).

Serving: binary 2b666cc74334abec

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	27.9 (27.9)	38.0 (38.0)	33.2 (33.2)	30.6 (30.6)	0.18-0.23 s
4	48.9 (12.3)	66.0 (16.7)	64.8 (16.2)	63.7 (16.1)	0.79-0.86 s
8	60.3 (7.6)	75.9 (9.5)	82.2 (10.3)	76.3 (9.4)	1.48-1.73 s
16	123.6 (7.6)	155.5 (9.6)	149.5 (9.0)	145.0 (9.1)	8.50-17.20 s

Suite-together

check	passed
concurrent	12/12
drafts	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 21:38	glm-streams	done	38.0	155.5	8.50-17.20 s	24/24
2026-10-16 21:24	glm-vision	done				22/22
2026-10-16 21:03	glm-vision	done				4/6
2026-10-16 18:10	glm-quality	done	38.2			

GLM-5.3-Flash EXL3 4bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-gb10-port16: GLM-5.3-Flash EXL3 4bpw, families/glm with 4096-row index rings, --parallel 16, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA: families/glm + index rings (16 streams at 40960); served as GLM-5.3-Flash-EXL3. Latest: run 20261017-062916-glm-streams (2026-10-17 06:29, plan glm-streams, done).

Serving: binary 142bc21ac97bc065

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	28.0 (28.0)	38.6 (38.6)	33.5 (33.5)	30.7 (30.7)	0.18-0.23 s
4	51.3 (12.9)	65.5 (16.3)	65.7 (16.5)	66.0 (16.5)	0.67-0.81 s
8	64.3 (8.1)	82.3 (10.3)	83.2 (10.4)	81.8 (10.2)	1.41-1.68 s
16	86.4 (5.4)	100.6 (6.3)	100.2 (6.1)	101.5 (6.2)	3.23-3.72 s

Suite-together

check	passed
concurrent	12/12
drafts	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 06:29	glm-streams	done	38.6	100.6	3.23-3.72 s	24/24
2026-10-17 05:56	glm-streams	done	39.0	90.8	3.20-3.67 s	24/24

GLM-5.3-Flash, EXL3 4 bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-sparks: GLM-5.3-Flash EXL3 4 bpw, TensorFold native, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA; served as GLM-5.3-Flash-EXL3. Latest: run evidence/2026-10-14 (2026-10-14 00:00, plan published evidence, published).

Ruler-shipped-14fb445e-c1-sampled

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	45.1 (45.2)	64.6 (65.4)	56.5 (56.2)	57.0 (54.9)	0.18-0.25 s

Ruler-shipped-14fb445e-c1

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	46.5 (45.4)	68.7 (69.4)	57.0 (56.4)	58.8 (55.3)	0.18-0.24 s

Ruler-shipped-14fb445e-c16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
16	144.0 (8.9)	198.3 (12.1)	180.1 (11.3)	208.4 (12.5)	0.91-1.19 s

Ruler-shipped-14fb445e-c8

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	110.1 (13.6)	152.5 (18.6)	138.7 (17.5)	157.0 (18.7)	0.49-0.63 s

Ruler-shipped-f95f2f70-c1-sampled

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	45.4 (45.4)	65.8 (67.0)	56.4 (55.9)	58.1 (55.3)	0.17-0.23 s

Ruler-shipped-f95f2f70-c1

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	47.5 (46.5)	69.5 (69.9)	58.2 (56.9)	60.2 (56.8)	0.16-0.23 s

Ruler-shipped-f95f2f70-c16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
16	144.8 (8.9)	202.2 (12.3)	181.4 (11.4)	207.6 (12.5)	0.85-1.14 s

Ruler-shipped-f95f2f70-c8

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	111.1 (13.7)	154.1 (18.7)	139.5 (17.2)	157.3 (18.8)	0.45-0.60 s

Ablit/suite

check	passed
gsm8k	246/250
mmlu	292/342
humaneval	159/164
ifeval	118/150
tools	59/60
json	30/30
mgsm	221/240

Suite-ab

check	passed
gsm8k	244/250
mmlu	292/342
humaneval	156/164
ifeval	118/150
tools	59/60
json	30/30
mgsm	223/240

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-14 00:00	published evidence	published	68.7			1125/1236
2026-10-13 00:00	published evidence	published	1083.6			1874/2172
2026-10-12 00:00	published evidence	published	62.8	169.5	0.45-1.19 s	

GLM-5.3-Flash, MLX 4-bit on 1x Apple M5 Ultra (256 GB unified), macOS

Target glm53-m5: GLM-5.3-Flash MLX 4-bit, TensorFold native on one M5 Ultra Mac (one stream, greedy). Engine: TensorFold native, Metal; served as GLM-5.3-Flash-MLX-4bit. Latest: run 20261016-082326-glm-gate-novision (2026-10-16 08:23, plan glm-gate-novision, done).

Serving: engine_commit e8b60549; binary_sha256 d90e44c9818187ae4bc61d06312989cd796033b5dea6e0fc02164fe2730660fe

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	74.4 (74.4)	78.3 (78.3)	77.5 (77.5)	86.5 (86.5)	0.11-0.15 s

Suite

check	passed
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2
max-tokens	3/3
tools	57/60
json	30/30
gsm8k	246/250
mmlu	290/342

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 08:23	glm-gate-novision	done	78.3			677/736
2026-10-16 08:04	glm-gate-novision	done	77.6			679/736
2026-10-16 00:35	glm-gate	done	78.7			689/748
2026-10-10 00:00	published evidence	published	103.7			36/36

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-10 00:00	published evidence	published				1128/1213
2026-10-10 00:00	published evidence	published				1137/1213
2026-10-10 00:00	published evidence	published				1079/1213
2026-10-09 00:00	published evidence	published	104.6			
2026-10-09 00:00	published evidence	published				349/403
2026-10-09 00:00	published evidence	published				42/42

NVIDIA Nemotron 3.5 Bolt 74B-A9B (24 GB: IQ2_XXS experts, Q8_0, NVFP4 Mamba) + vision on 1x NVIDIA RT

Target nemotron-bolt-5090: Nemotron 3.5 Bolt 74B-A9B ModelOpt IQ2_XXS/Q8_0/NVFP4, TensorFold native CUDA on one RTX 5090 (262144, two streams, vision, host prompt cache). Engine: TensorFold native, CUDA; served as nemotron-bolt. Latest: run 20261015-175621-full (2026-10-15 17:56, plan full, done).

Serving: recipe_commit f546a71; engine_commit b2ec378d; served_model {"object": "list", "data": [{"id": "nemotron-bolt", "object": "model", "created": 17920869

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	149.7 (149.7)	149.6 (149.6)	150.1 (150.1)	149.7 (149.7)	0.03-0.03 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	1.011	0.0		6.75	217

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	149.3 (149.3)	149.2 (149.2)	149.7 (149.7)	149.3 (149.3)	0.03-0.03 s

Engine rounds (ruler-after):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	1.007	0.0		6.74	1781

Suite

check	passed
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2

check	passed
max-tokens	3/3
tools	58/60
json	30/30
ifeval	120/150
gsm8k	241/250
mgsm	190/240
mmlu	279/342
humaneval	153/164
repetition	8/8
long-context	55/55
vision	12/12
soak	60/60
snapshot	12/12

Checks

check	result
memory-idle	measured: gpu: 28716 MiB, 32607 MiB
structured	pass: 18/18 obey; a bad regex -> 400
sampling	pass: 9/9 seeded replies equal on repeat; 0 errors
turns	pass: resumed == again == cold: True; cached 5832 of 5865; ttft resumed 0.053s, cold 0.134s
capacity	pass: 6/6 answered, 0 refused, 0 lost (no reply); a short request after: 200

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-15 17:56	full	done	149.6			1272/1437

Ornith-1.5-122B, 4-bit MLX affine (g64) on 1x Apple M5 Ultra (256 GB unified), macOS

Target ornith-15-122b-m5: Ornith-1.5-122B, 4-bit MLX affine with images, vllm-mlx on one M5 Ultra. Engine: vllm-mlx 0.6.0 (mlx-vlm 0.7.9, mlx-lm 0.31.3, mlx 0.32.4); served as Ornith-1.5-122B. Latest: run 20261020-111724-mlx-speed (2026-10-20 11:17, plan mlx-speed, done).

Serving: stack vllm-mlx 0.6.0, mlx-vlm 0.7.9, mlx-lm 0.31.3, mlx 0.32.4; served_model {"object":"list","data":[{"id":"Ornith-1.5-122B","object":"model","created":1792513044,"ow; checkpoint {'dir': 'Ornith-1.5-122B-MLX-4bit', 'bits': 4, 'group_size': 64, 'mode': 'affine', 'revisi; chip Apple M5 Ultra; 274877906944; suite_commit 277c842a1481

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	66.2 (66.2)	66.6 (66.6)	66.4 (66.4)	66.6 (66.6)	0.08-0.11 s
2	113.4 (56.7)	113.2 (56.6)	113.2 (56.6)	114.0 (57.0)	0.14-0.19 s
4	175.9 (44.0)	178.1 (44.9)	177.6 (44.1)	180.4 (44.9)	0.25-0.35 s

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 95%; total = 7168.00M us

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-20 11:17	mlx-speed	done	66.6			

Ornith-1.5-122B, 4-bit MLX affine (g64) on 1x Apple M5 Ultra (256 GB unified), macOS

Target ornith-15-122b-native-m5: Ornith-1.5-122B, 4-bit MLX affine, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as Ornith-1.5-122B. Latest: run 20261020-133912-mlx-speed (2026-10-20 13:39, plan mlx-speed, done).

Serving: engine_commit 32dad79b; binary_sha256 5dcb999c71614342b4d9056e76ec2d00befb6d581f6961bcc088d5a7cd2a3bd1; served_model {"object": "list", "data": [{"id": "Ornith-1.5-122B", "object": "model", "created": 179252; chip Apple M5 Ultra; 274877906944; suite_commit 277c842a1481

Ruler-streams

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	85.0 (85.0)	84.8 (84.8)	84.6 (84.6)	84.9 (84.9)	0.05-0.07 s
2	139.2 (69.6)	143.6 (71.8)	146.2 (73.1)	141.7 (70.8)	0.07-0.10 s
4	218.7 (54.5)	223.6 (56.1)	238.1 (61.0)	227.3 (56.8)	0.12-0.16 s

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 66%; total = 5120.00M us

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-20 13:39	mlx-speed	done	84.8			
2026-10-20 13:36	mlx-speed	done	84.1			
2026-10-20 13:31	mlx-speed	done	85.3			
2026-10-20 13:29	mlx-speed	done	84.2			
2026-10-20 13:19	mlx-speed	done	84.6			
2026-10-20 12:56	mlx-speed	done	79.3			
2026-10-20 12:49	mlx-speed	done	85.1			

Qwen3.8-27B, MLX 4-bit + DFlash2 on 1x NVIDIA RTX 5090 (32 GB)

Target qwen38-27b-5090: Qwen3.8-27B MLX 4-bit + DFlash2, TensorFold native CUDA on one RTX 5090 (fp8 KV, 262k, one stream, vision). Engine: TensorFold native, CUDA; served as qwen38-27b. Latest: run 20261016-111449-quick (2026-10-16 11:14, plan quick, done).

Serving: recipe_commit 160a2c0; engine_commit bfcc84ab; served_model {"object": "list", "data": [{"id": "qwen38-27b", "object": "model", "created": 1792149307,

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	240.7 (240.7)	459.0 (459.0)	309.5 (309.5)	313.5 (313.5)	0.10-0.10 s

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	239.8 (239.8)	457.6 (457.6)	308.9 (308.9)	313.4 (313.4)	0.10-0.10 s

Suite

check	passed
soak	60/60
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	7/8
stop	3/3
thinking	10/10
copy	2/2
max-tokens	3/3

Checks

check	result
memory-idle	measured: gpu: 20958 MiB, 32607 MiB
sampling	pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 11:14	quick	done	459.0			113/114
2026-10-14 00:00	published evidence	published				
2026-10-13 00:00	published evidence	published				
2026-10-12 00:00	published evidence	published				1229/1437
2026-10-12 00:00	published evidence	published				1289/1437
2026-10-12 00:00	published evidence	published				
2026-10-12 00:00	published evidence	published				

Qwen3.8-Flash-Next, MLX 6-bit on 1x Apple M5 Ultra (256 GB unified), macOS

Target qwen38-flashnext-mlx6bit-m5: Qwen3.8-Flash-Next MLX 6-bit, TensorFold native on one M5 Ultra. Engine: TensorFold native, Metal; served as Qwen3.8-Flash-Next-MLX-6bit. Latest: run evidence/2026-10-11-suite-rest (2026-10-11 00:00, plan published evidence, published).

Suite

check	passed
ifeval	134/150
gsm8k	242/250
mgsm	225/240
mmlu	282/342
repetition	8/8
long-context	55/55
concurrent	12/12
snapshot	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-11 00:00	published evidence	published				970/1069
2026-10-11 00:00	published evidence	published	205.9			24/24
2026-10-11 00:00	published evidence	published				157/164
2026-10-11 00:00	published evidence	published	199.8			
2026-10-11 00:00	published evidence	published				131/132
2026-10-11 00:00	published evidence	published	206.2			

Qwen3.8-Flash-Next, 6-bit with vision on 1x Apple M5 Ultra (256 GB unified), macOS

Target qwen38-flashnext-vision-m5: Qwen3.8-Flash-Next 6-bit MTP with images and video, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as Qwen3.8-Flash-Next-MLX-6bit. Latest: run 20261017-120921-fn-speed (2026-10-17 12:09, plan fn-speed, done).

Serving: engine_commit 394d2e35; binary_sha256 43a9c01ac3eee31998def08595f5e4fc6d29819d46799a8ac01b7cdffd72b668; served_model {"object": "list", "data": [{"id": "Qwen3.8-Flash-Next-MLX-6bit", "object": "model", "crea; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	142.4 (142.4)	209.5 (209.5)	173.2 (173.2)	206.5 (206.5)	0.11-0.13 s
2	197.6 (98.8)	261.9 (130.9)	221.5 (110.7)	271.5 (135.8)	0.17-0.19 s

Suite

check	passed
drafts	12/12
concurrent	12/12

Checks

check	result
structured	pass: 18/18 obey; a bad regex -> 400

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 12:09	fn-speed	done	209.5			24/24
2026-10-17 12:06	fn-speed	done	209.8			24/24
2026-10-17 11:34	fn-1m	done	197.4			24/24
2026-10-17 11:31	fn-media	done	208.9			24/24
2026-10-17 08:04	fn-speed	done	206.1			24/24
2026-10-17 08:02	fn-speed	done	204.3			24/24
2026-10-17 05:58	fn-speed	done	208.0			24/24
2026-10-17 01:00	fn-speed	done	202.8			24/24

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 00:57	fn-speed	done	207.3			24/24
2026-10-16 19:29	fn-speed	done	201.8			24/24
2026-10-16 19:26	fn-speed	done	207.3			24/24
2026-10-12 00:00	published evidence	published	174.1			
2026-10-12 00:00	published evidence	published	202.7			
2026-10-12 00:00	published evidence	published	176.6			
2026-10-11 00:00	published evidence	published	205.6			131/132
2026-10-11 00:00	published evidence	published	178.9			513/544

Qwen3.8-Flash-Next INT4-Mixed (AutoRound) on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 ov

Target qwen38-int4mixed: Qwen3.8-Flash-Next INT4-Mixed, TensorFold native, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA; served as Qwen3.8-Flash-Next-INT4-Mixed. Latest: run 20261017-072502-rebase (2026-10-17 07:25, plan rebase, done).

Serving: patch_sha 47c78d3f99593702c9e8a0baa64ec32c937520930186c78549004f411cd5f731; image tf-qwen38-int4mixed:native-v1.0.12 sha256:276fe2ba4ed3e778ec6f55ffad1430897b9ff439aff8352b; server_args --context 1048576 --no-thinking --parallel 16 --vision-urls --vision

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	78.0 (78.0)	127.0 (127.0)	102.1 (102.1)	130.5 (130.5)	0.03-0.04 s
16	304.2 (18.9)	516.3 (31.0)	391.6 (24.0)	524.9 (31.7)	0.45-0.54 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.621	4.042	66%	33.09	12
16	3.212	2.921	76%	121.71	193

Suite

check	passed
soak	60/60
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2
max-tokens	3/3

Suite-guided

check	passed
json-guided	30/30
tools	60/60

Suite-thinking

check	passed
gsm8k	234/250

Checks

check	result
memory-idle	measured: head 22.6 GiB available; worker 25.3 GiB available
structured	pass: 18/18 obey; a bad regex -> 400
sampling	pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors
turns	pass: resumed == again == cold: True; cached 5575 of 5642; ttft resumed 0.105s, cold 0.035s
thinking	pass: thinking on: 10/10 right, 10/10 with reasoning_content, 0 leaks, 0 errors

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-17 07:25	rebase	done	127.0	516.3	0.45-0.54 s	114/114
2026-10-15 17:13	full	done	127.0	509.8	0.44-0.55 s	1321/1437
2026-10-15 00:00	published evidence	published	123.4	472.9	0.52-0.74 s	1321/1437
2026-10-15 00:00	published evidence	published				1321/1437
2026-10-11 00:00	published evidence	published	120.1	507.2	0.54-0.62 s	1321/1437
2026-10-11 00:00	published evidence	published	120.4	507.1	0.50-0.59 s	
2026-10-10 09:45	quick	done	105.3			54/54
2026-10-10 00:00	published evidence	published	122.0	510.3	0.40-0.49 s	114/114
2026-10-10 00:00	published evidence	published	125.9	510.9	0.44-0.49 s	1261/1377
2026-10-10 00:00	published evidence	published	125.7	506.7	0.42-0.49 s	54/54
2026-10-10 00:00	published evidence	published	106.1	506.1	0.42-0.51 s	54/54

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-10 00:00	published evidence	published	110.0	511.8	0.41-0.55 s	1261/1377
2026-10-10 00:00	published evidence	published	105.8	505.5	0.41-0.46 s	1261/1377
2026-10-09 00:00	published evidence	published	105.2	412.2	0.78-0.98 s	1260/1377
2026-10-09 00:00	published evidence	published				1256/1377
2026-10-08 00:00	published evidence	published	107.1	519.4	0.44-0.56 s	
2026-10-08 00:00	published evidence	published	107.7	474.0	0.52-0.66 s	145/145
2026-10-08 00:00	published evidence	published	107.6	442.6	0.44-0.58 s	36/36
2026-10-08 00:00	published evidence	published	107.6	441.7	0.45-0.58 s	
2026-10-07 00:00	published evidence	published				12/12

Step-5-Preview, MLX 2-bit experts (8-bit rest) + MTP on 1x Apple M5 Ultra (256 GB unified), macOS

Target step5-m5: Step-5-Preview 2-bit (experts 2/128, rest 8/64, MTP, images), TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as step5-preview. Latest: run 20261016-102144-pr903-final (2026-10-16 10:21, plan pr903-final, done).

Serving: engine_commit e2bb58db; binary_sha256 61059a383f74fd363f17527526fdc7de028fe624d40e59e3238d20e4925bfef5; served_model {"object": "list", "data": [{"id": "step5-preview", "object": "model", "created": 17921641; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	38.5 (38.5)	50.4 (50.4)	40.5 (40.5)	47.9 (47.9)	0.24-0.28 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	2.381	2.072	67%	53.72	12

Suite

check	passed
vision	12/12
drafts	12/12
concurrent	12/12

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 27%; total = 4096.00M us
turns	pass: resumed == again == cold: True; cached 4514 of 4618; ttft resumed 0.353s, cold 3.21s

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-16 10:21	pr903-final	done	50.4			36/36
2026-10-16 09:23	pr903	done	50.4			156/156
2026-10-15 19:44	one-stream	done	50.9			220/232

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-15 19:28	quick	done	50.5			114/114
2026-10-15 18:52	quick	done	50.6			114/114
2026-10-15 13:43	one-stream	done	50.6			220/232
2026-10-15 12:43	quick	done	50.6			

Served earlier, numbers in their READMEs only

These recipes predate the ruler's JSON, so their measurements are not in the ledger; their READMEs hold them.

model	hardware	engine	recipe
Qwen3.8-Flash-Next, turboderp EXL3 4.05 bpw	2x NVIDIA DGX Spark (GB10), TP=2	TensorFold 0.6.6 (Python) with an EXL3 patch	github.com/BobClawblaw/qwen38-flashnext-exl3-2x-dgx-sparks
Qwen3.8-Flash-Next (TensorFold 0.6.6 recipe)	2x NVIDIA DGX Spark (GB10), TP=2	TensorFold 0.6.6 (Python)	github.com/BobClawblaw/qwen38-flashnext-tensorfold-2x-dgx-sparks

Step-5-Preview on one M5 Ultra, final build: measured outside the harness

TensorFold native Metal, branch step5-native at engine 6e791da6: 2,048-row prompt chunks, the prompt cache and the idle keepalive's kernel tick. Greedy unless said; served with --parallel 2 --vision at the default context (227,025).

what	measured
prompt, 4k / 8k / 16k / 32k tokens	868 / 863 / 840 / 800 tok/s (a 4k prompt right after a pause was 580 before the keepalive fix)
212k-token chat question, one record's code of 10,500	2 of 3 right (records 3,000 and 9,800; missed 777), 376 s each, at least 19% of memory free
129k-token chat question	2 of 2 right (records 777 and 5,000), about 200 s each
MTP drafts against "draft": false, 200-400-token replies timed with their prompt	identical; code 25.8 to 41.2 tok/s, prose 27.0 to 36.6, math 26.9 to 48.8
prompt cache, an 11.6k-token chat sent again, then two more turns	15.3 s fresh, 1.8 s again, turns resumed at 11,601; replies identical to --prompt-cache-gib 0
prompt cache, images (1200 x 800 PNG, 688 tokens)	resent in 2.15 s instead of 3.8 s; replies identical to the cache off
a kept state that cannot sit beside its stream (2.4 GiB cache budget)	dropped before the pass; the prompt read again with the same reply
a 1-token reply after 2 / 6 / 30 / 120 s idle	0.17 / 0.18 / 0.18 / 0.19 s (about 3 s before the kernel tick and residency set)
two 18k/31k-token chats at once	both right

Step-5-Preview on one M5 Ultra, first build (512-row chunks): measured outside the harness

TensorFold native Metal, branch step5-native (engine 5ad7660c, images aa221b11), MLX 2-bit experts with the rest 8-bit, greedy, one stream. The harness's step5-m5 runs hold the ruler and the suite; these are the rest.

what	measured
prompt, 4k / 8k / 16k / 32k tokens	690 / 668 / 647 / 622 tok/s (the sparse indexer caps each row's keys past 4k)
131,016-token chat question about its first page	answered exactly, 264 s end to end, at least 23% of memory free
decode without drafts, 20 / 600 / 4,200 / 32,768 tokens of context	29.6 / 21.9 / 14.7 / 14.5 tok/s
MTP drafts against "draft": false (replies token-identical)	code 25.7 to 44.6 tok/s (70% accepted), prose 26.4 to 34.8 (56%), math 26.2 to 46.0 (75%)
an image, 1200 x 800 PNG (six 504-px crops and the overview, 692 prompt tokens)	2.1 s to a 4-token answer
an image, 600 x 400 PNG (overview only, 193 prompt tokens)	0.7 s